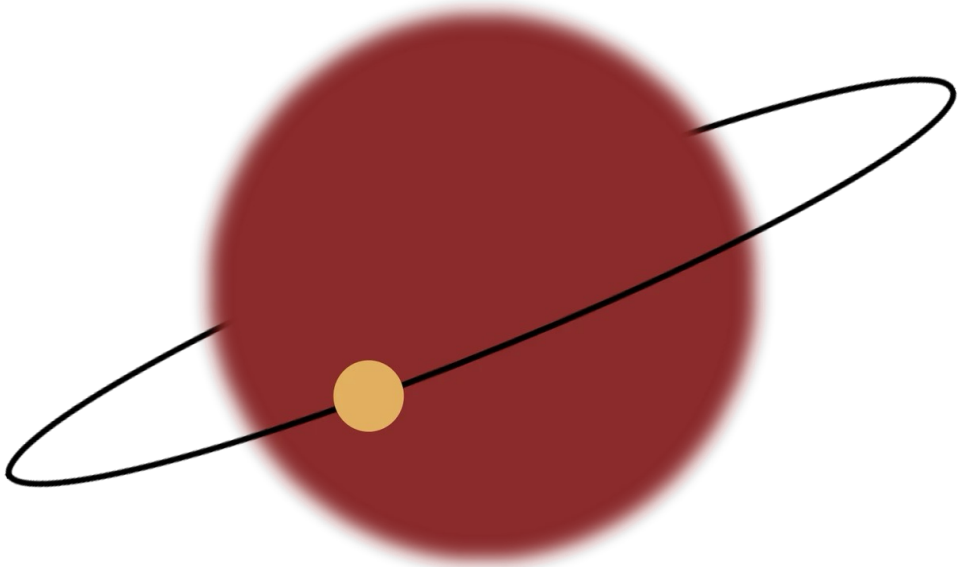
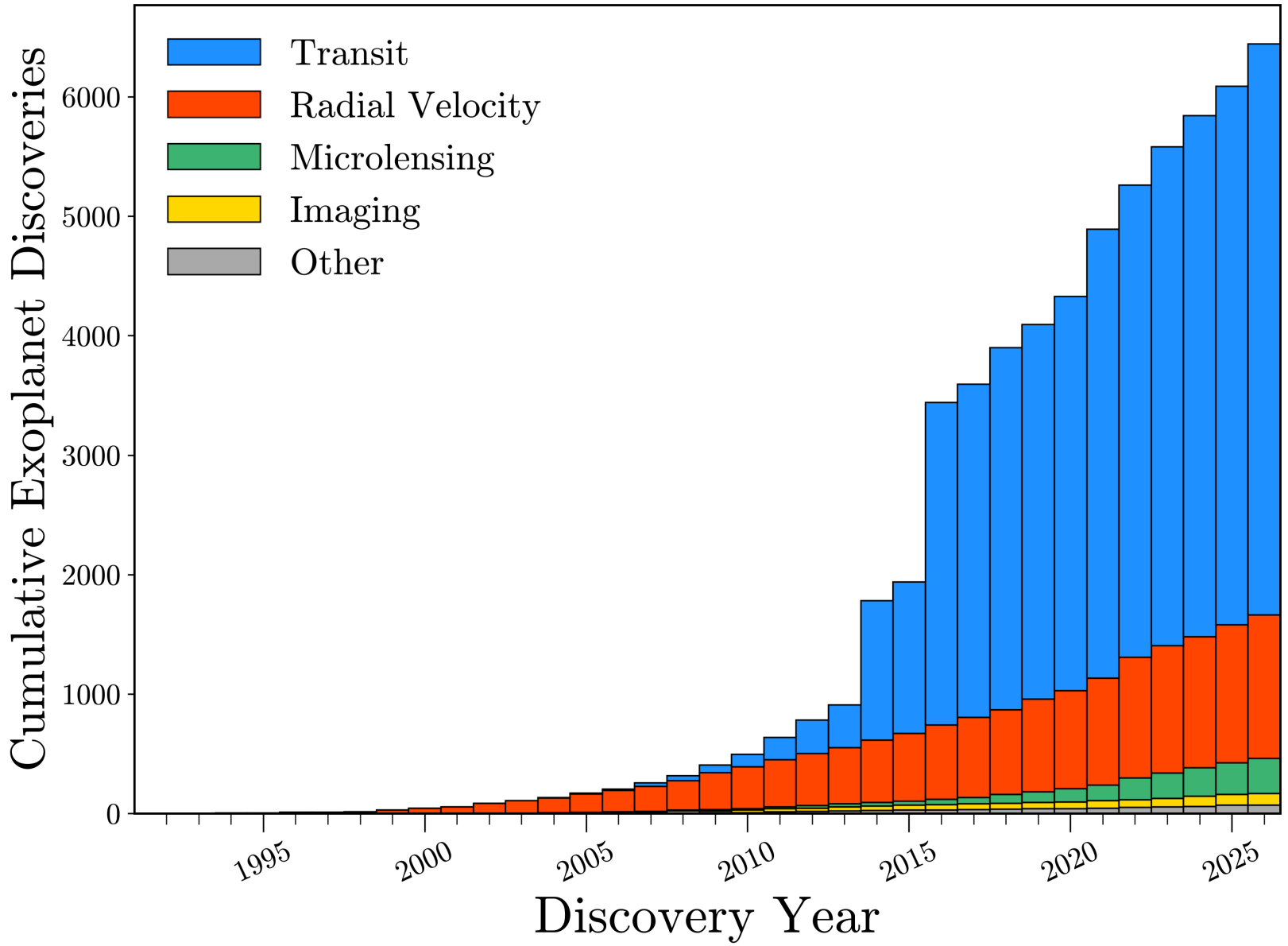


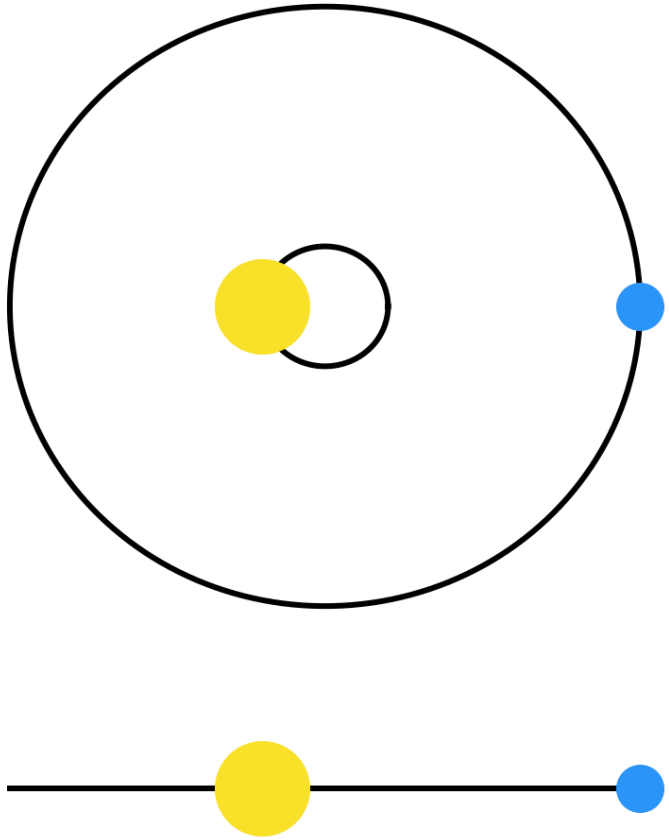
How to Find a Planet

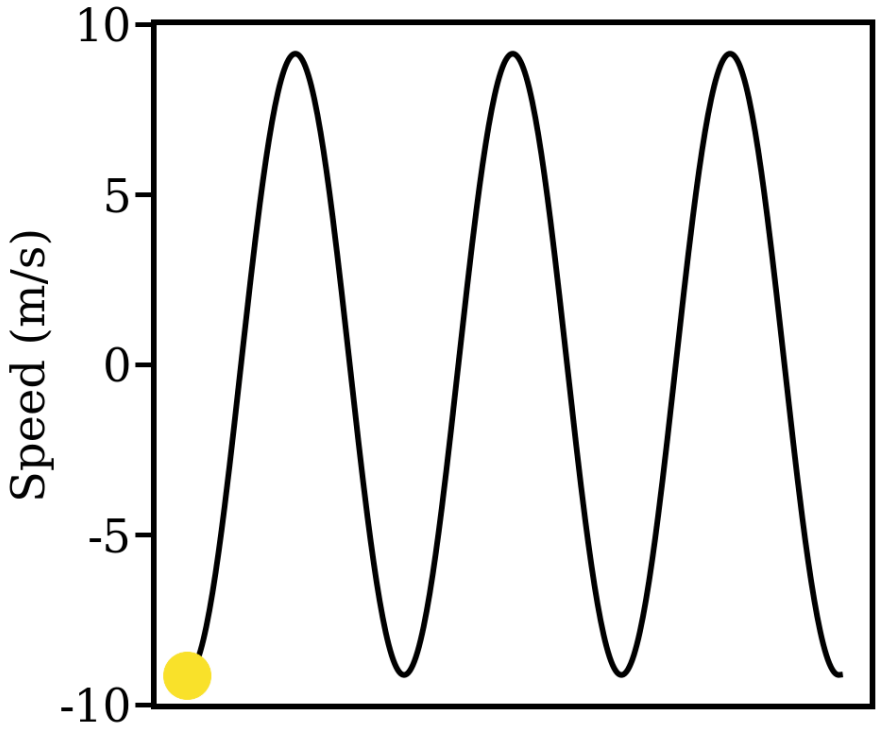
Dr. Nicholas Saunders
ASTR 2550, Yale University
16 October 2026

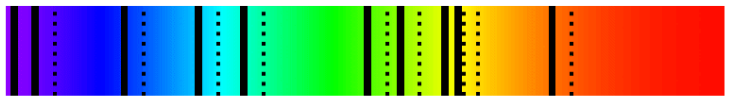

Transit
Radial Velocity
Microlensing
Imaging
Other
Cumulative Exoplanet Discoveries
0
1000
2000
3000
4000
5000
6000
1995
2000
2005
2010
2015
2020
2025
Discovery Year

The Radial Velocity Method

Alysa Obertas (@AstroAlysa)







The Radial Velocity Method

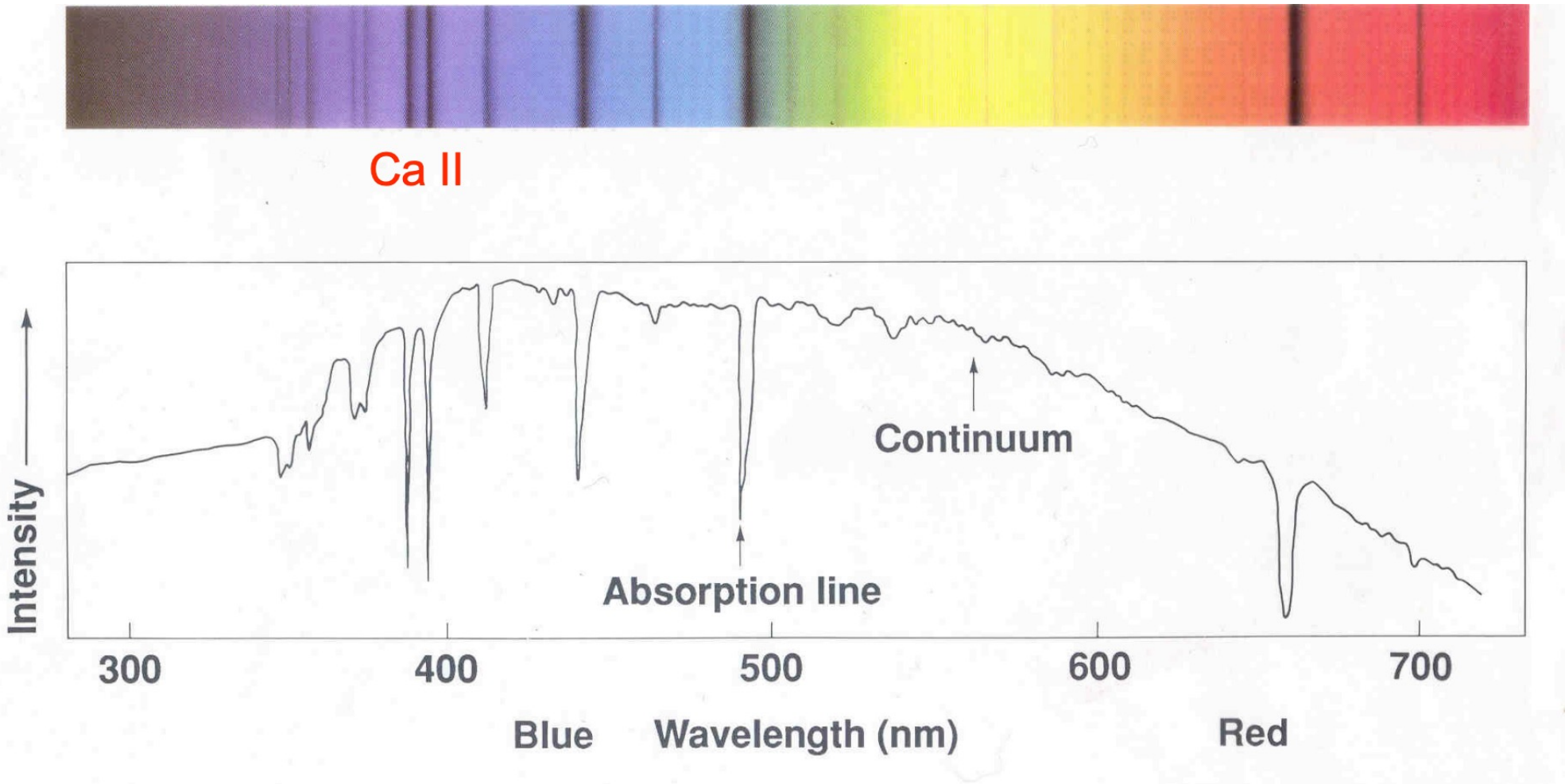
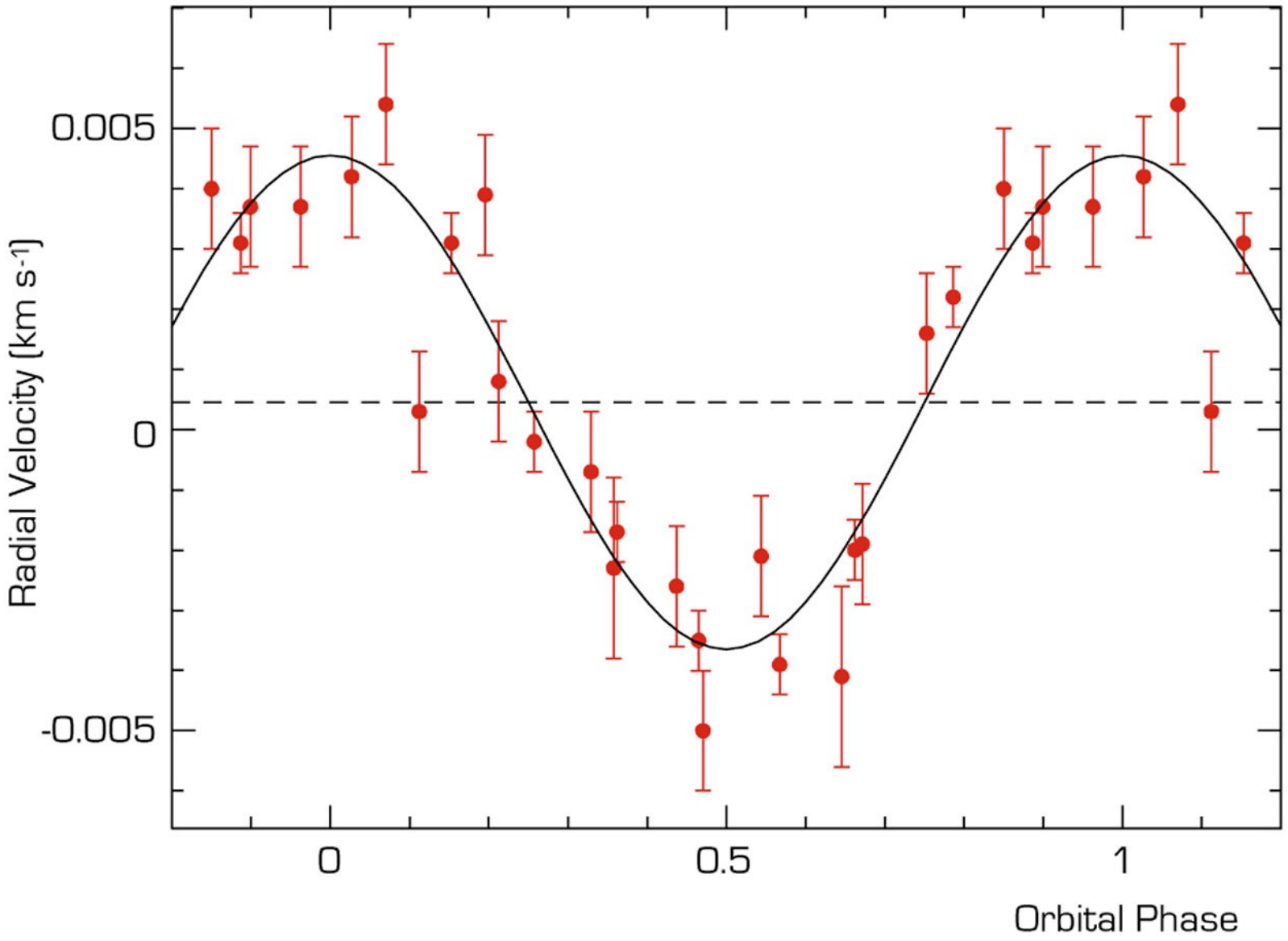


Image: Princeton

The Radial Velocity Method



$$K = \left(\frac{2\pi G}{M_\star^2 P}\right)^{1/3} \frac{1}{\sqrt{1-e^2}} M_p \sin i$$

- K — RV semi-amplitude
- G — gravitational constant
- $M_\star$ — stellar mass
- P — orbital period
- e — eccentricity
- M_p — planet mass
- i — orbital inclination

Image: ESO

The Radial Velocity Method

measuring shifts of spectral lines due to the star wobbling as the planet orbits

$$K = \left(\frac{2\pi G}{M_\star^2 P}\right)^{1/3} \frac{1}{\sqrt{1-e^2}} M_p \sin i$$

best at finding

small (terrestrial) **or** **large** (gas giant)

planets on

short (days) **or** **long** (years)

orbits.

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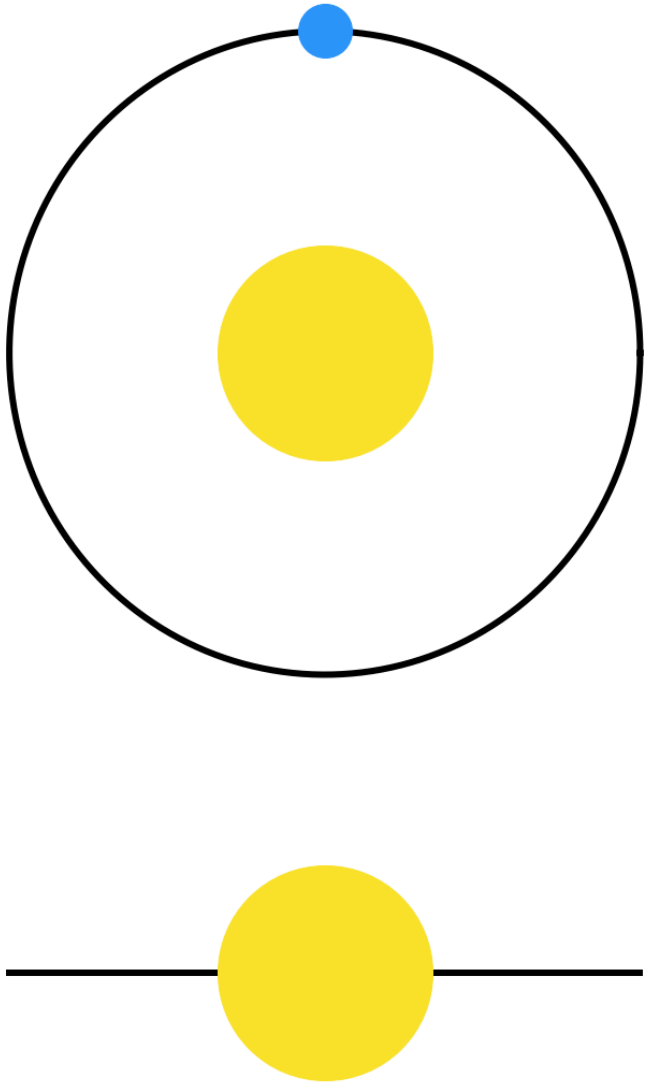
planets on

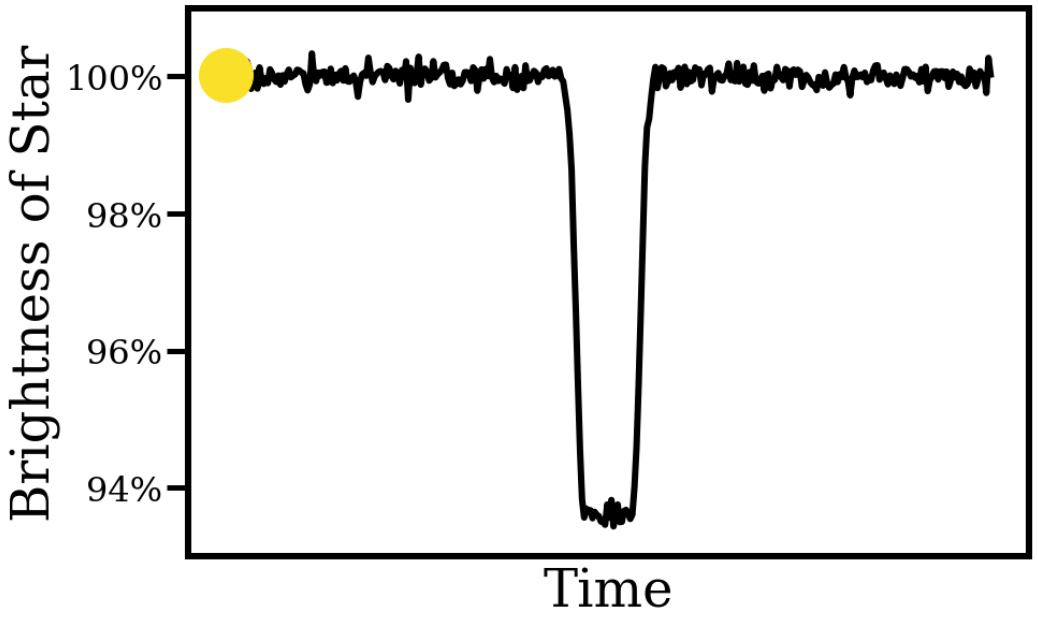
short (days) **or** **long** (years)

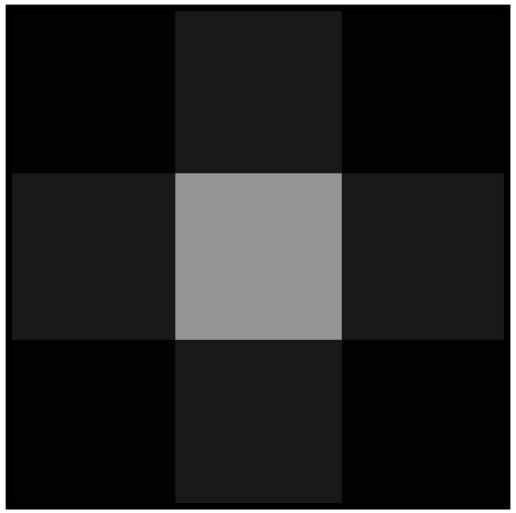
orbits.

The Transit Method

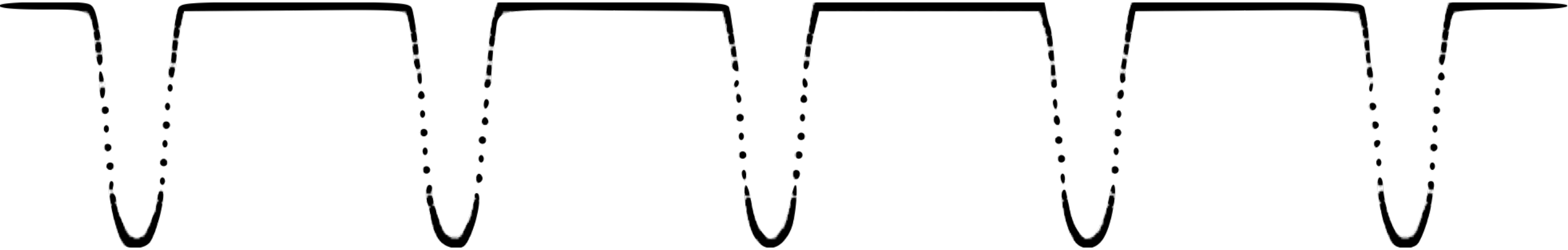
Alysa Obertas (@AstroAlysa)



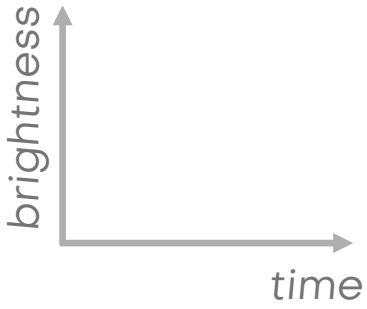




The Transit Method

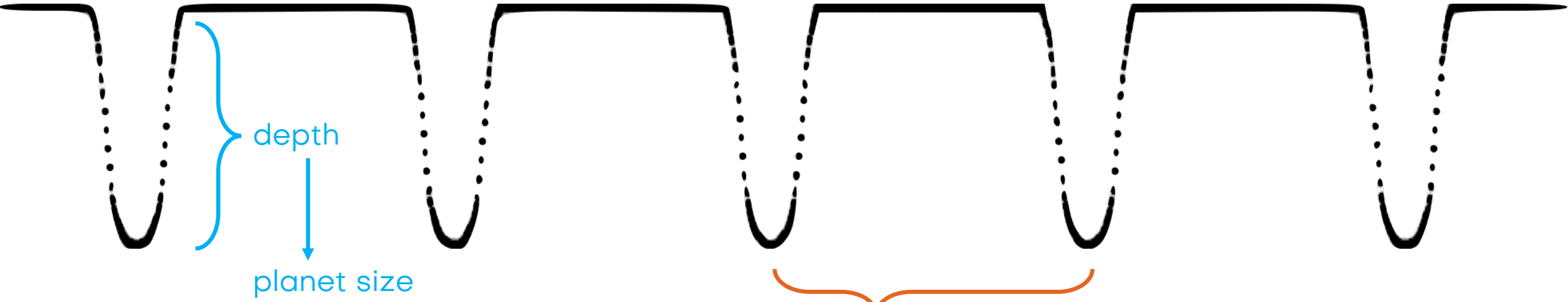


if you watch a star for a long time, you'll see multiple dips



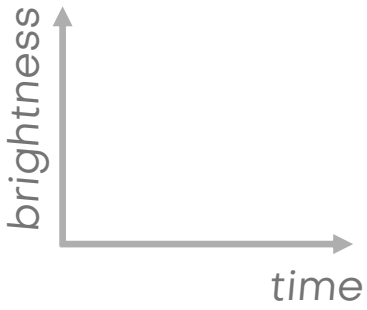
The Transit Method

$$\text{depth} = \left(\frac{R_p}{R_\star}\right)^2$$



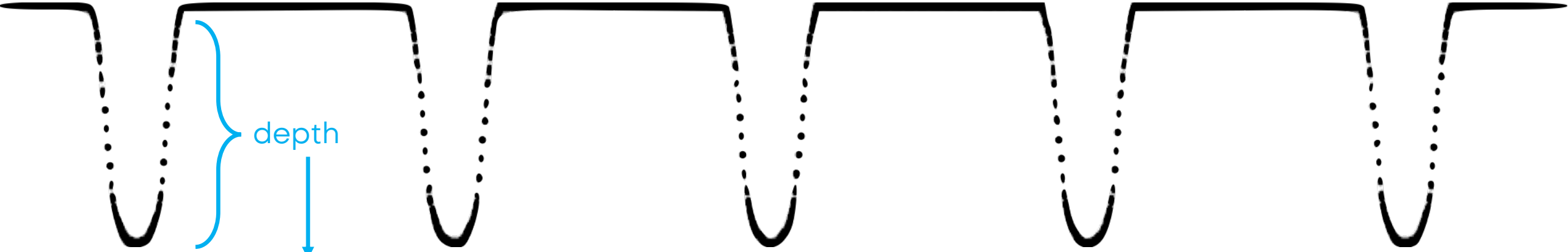
period → length of "year" → distance from star

$$P^2 = \left(\frac{4\pi^2}{GM}\right) a^3$$



The Transit Method

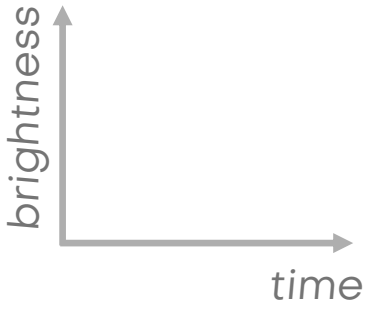
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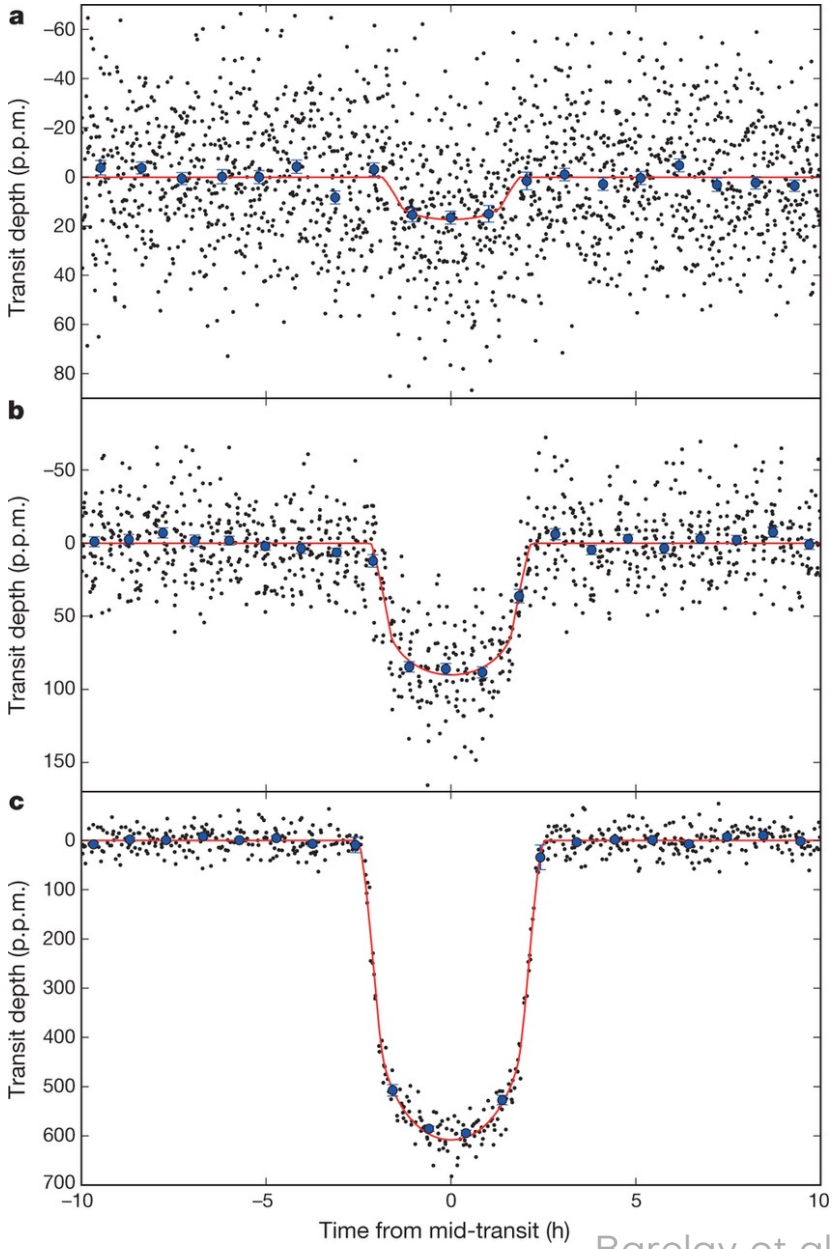
planet density

$$\rho_p = \frac{3M_p}{4\pi R_p^2}$$

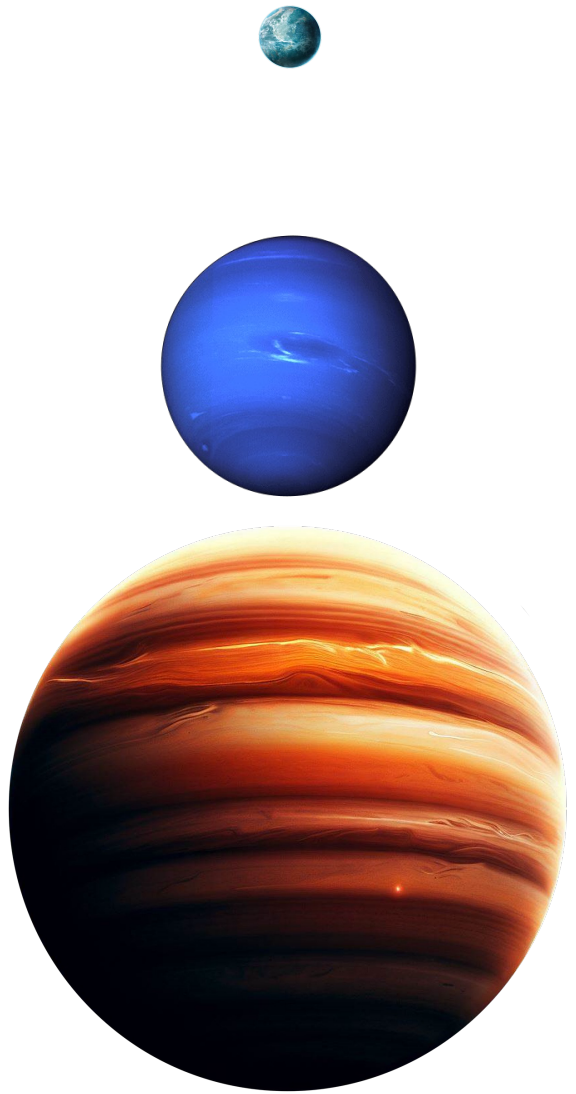
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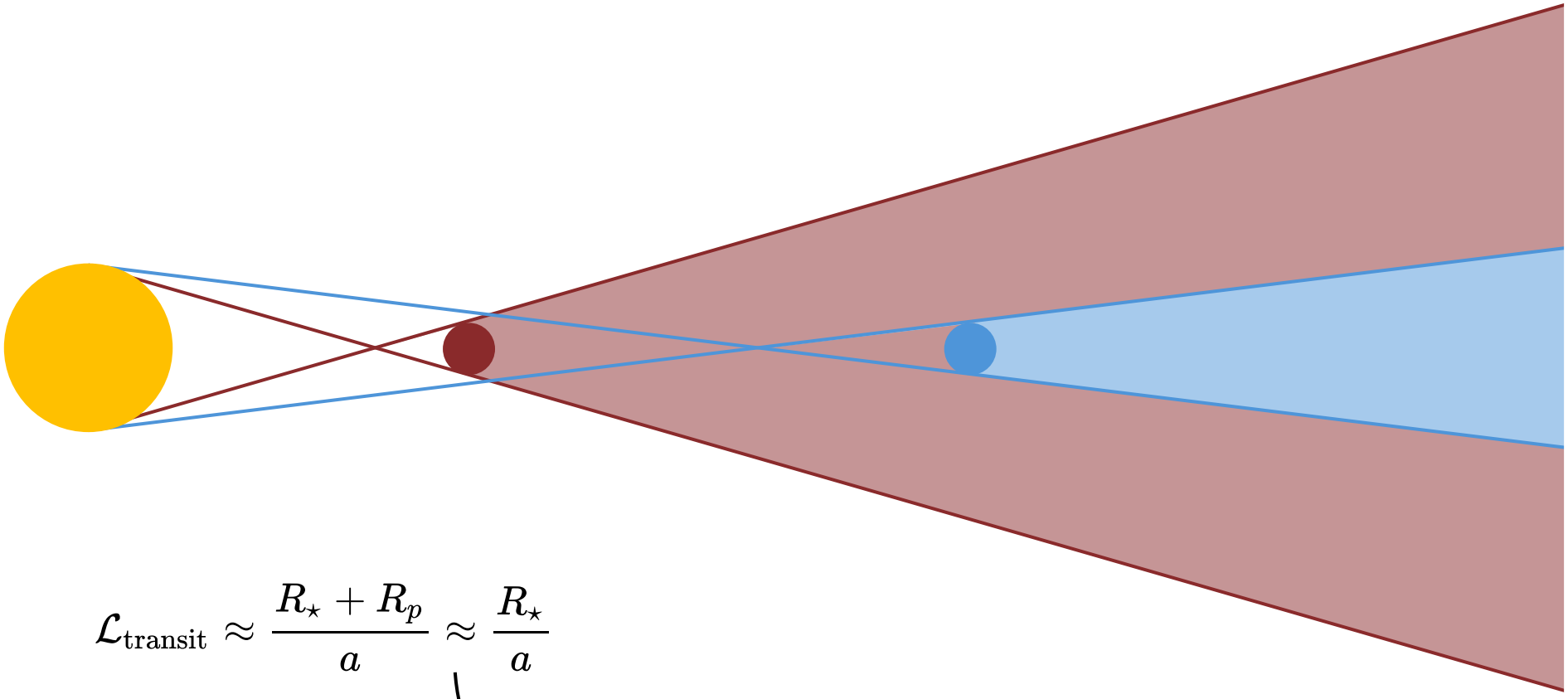
The Transit Method



Barclay et al. (2013)



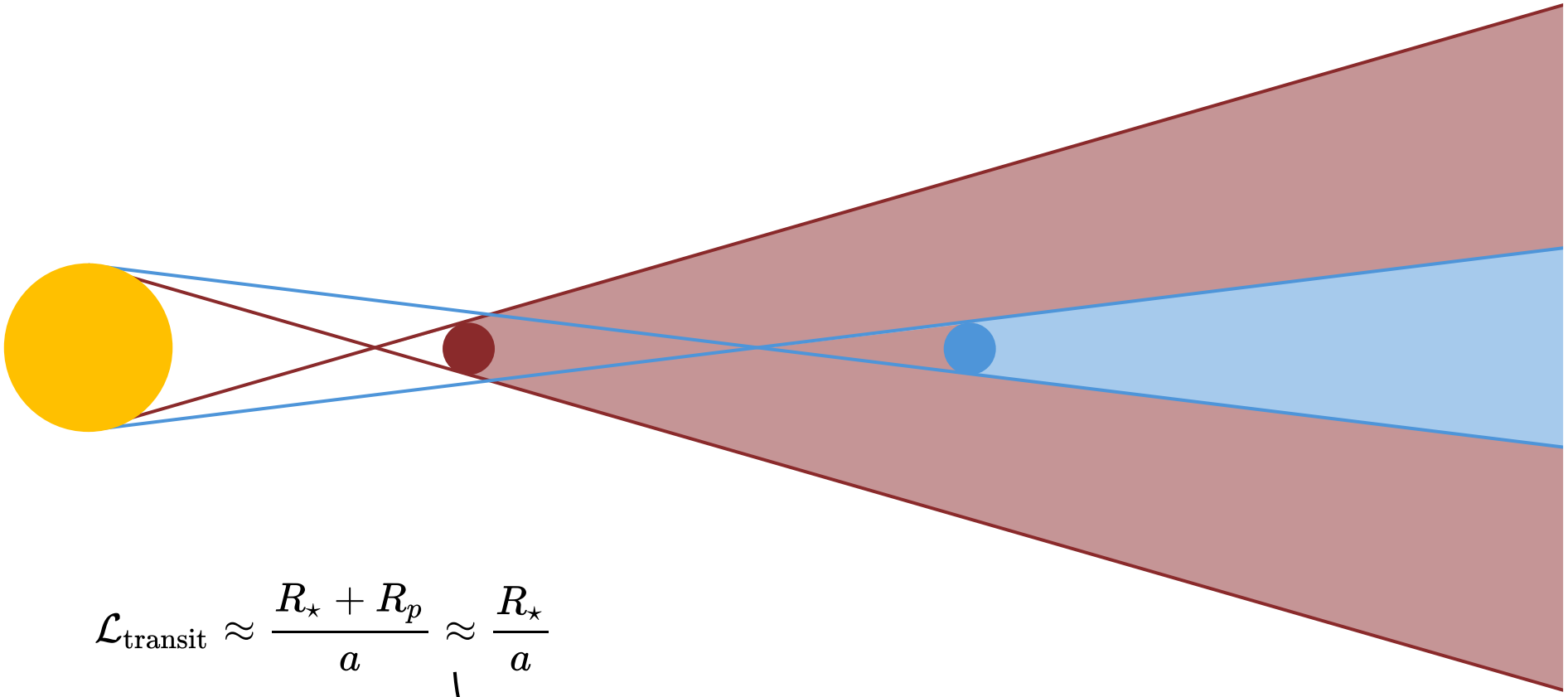
Geometric Transit Likelihood



$$\mathcal{L}_{\text{transit}} \approx \frac{R_\star + R_p}{a} \approx \frac{R_\star}{a}$$

assuming $R_p \ll R_\star$

Geometric Transit Likelihood



one planet!

two planets!

one planet!

no planet :(

$$\mathcal{L}_{\text{transit}} \approx \frac{R_\star + R_p}{a} \approx \frac{R_\star}{a}$$

assuming $R_p \ll R_\star$

The Transit Method

searching for periodic dips in the brightness of stars
caused by a planet blocking light once per orbit

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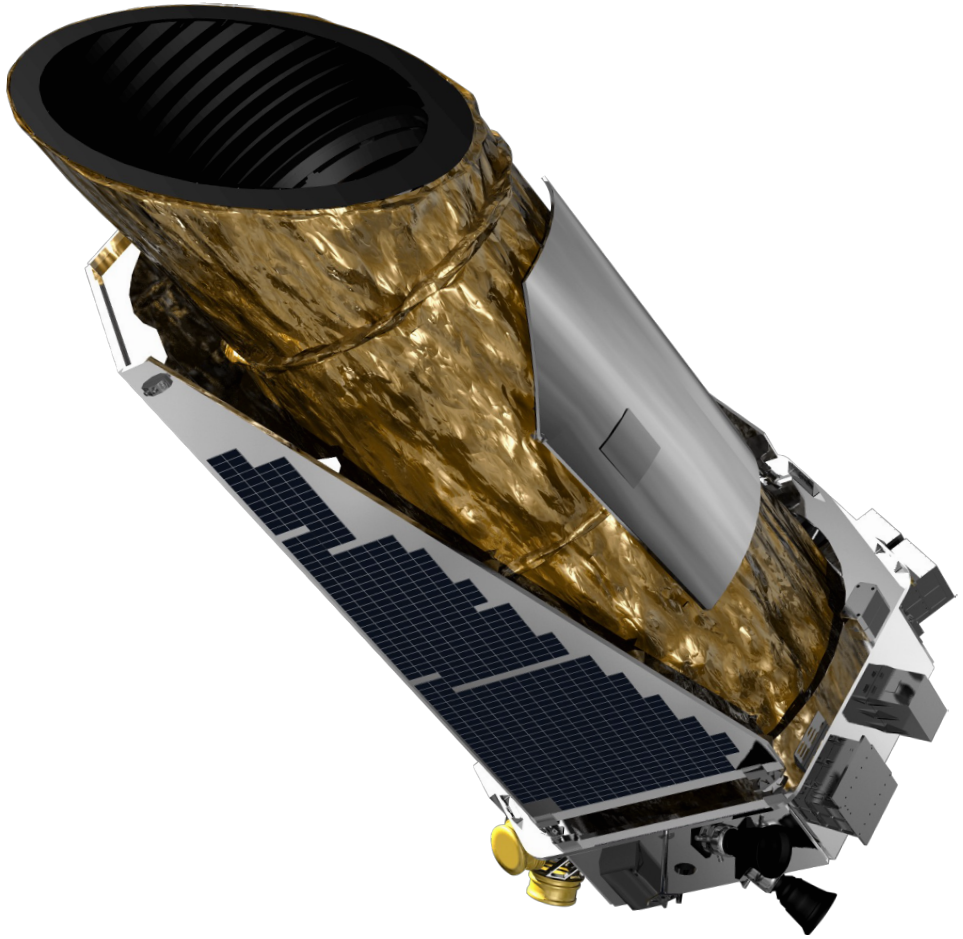
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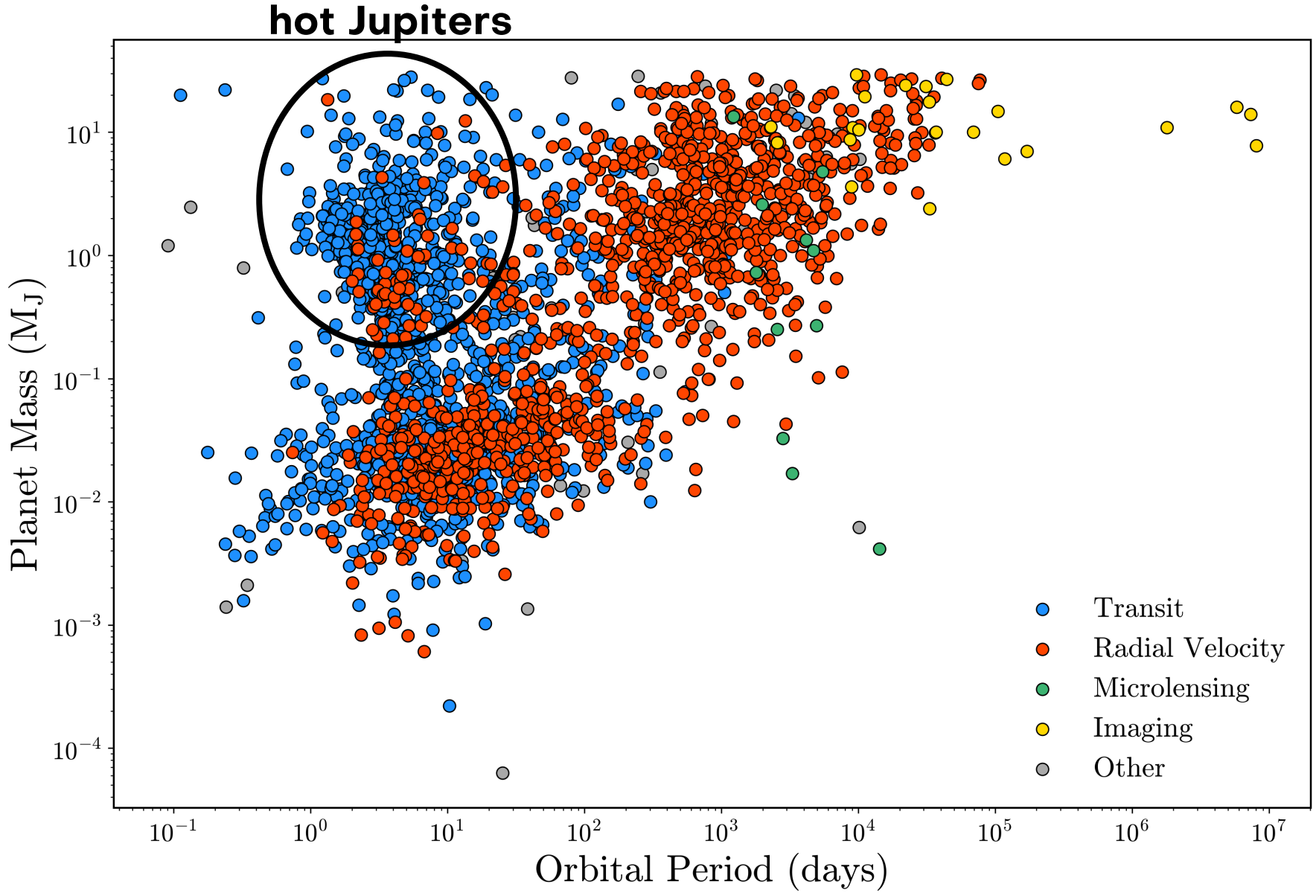
Kepler
2009 – 2018



TESS
2018 – present

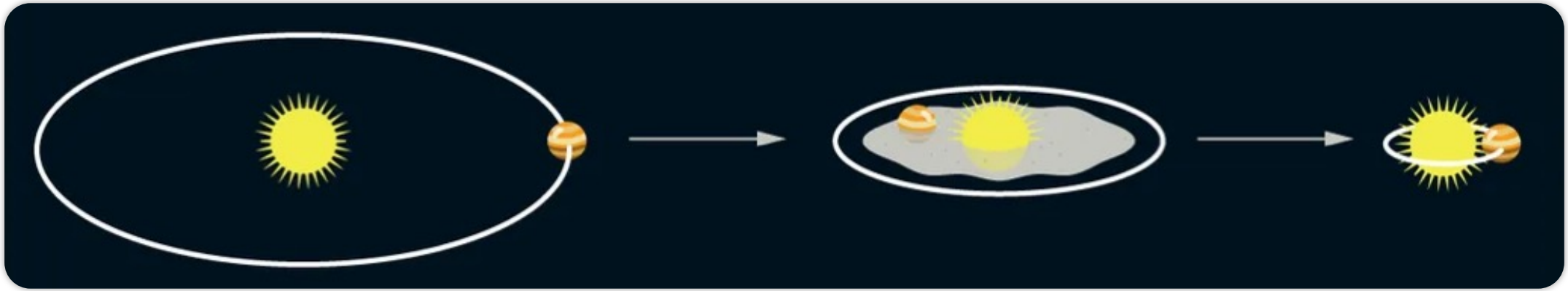


Images: NASA Goddard Space Flight Center

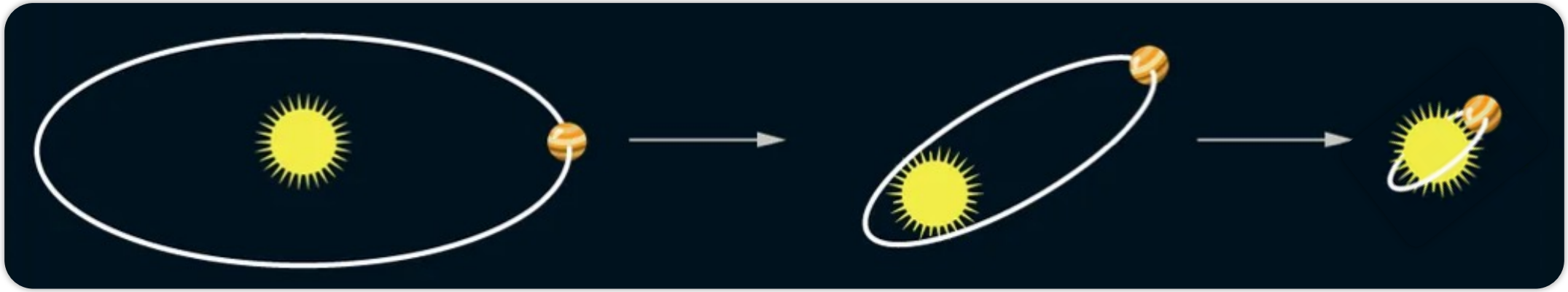

hot Jupiters
Planet Mass (M_J)
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
10^{5}
10^{6}
10^{7}
Orbital Period (days)
Transit
Radial Velocity
Microlensing
Imaging
Other

Hot Jupiter Formation Pathways

Disk-driven migration



High-eccentricity migration



Adapted from Dawson & Johnson (2018)

hot Jupiters are unlike anything we see in our solar system



Image: ESO / L. Calçada

hot Jupiters are unlike anything we see in our solar system

...so why do we find so many of them?

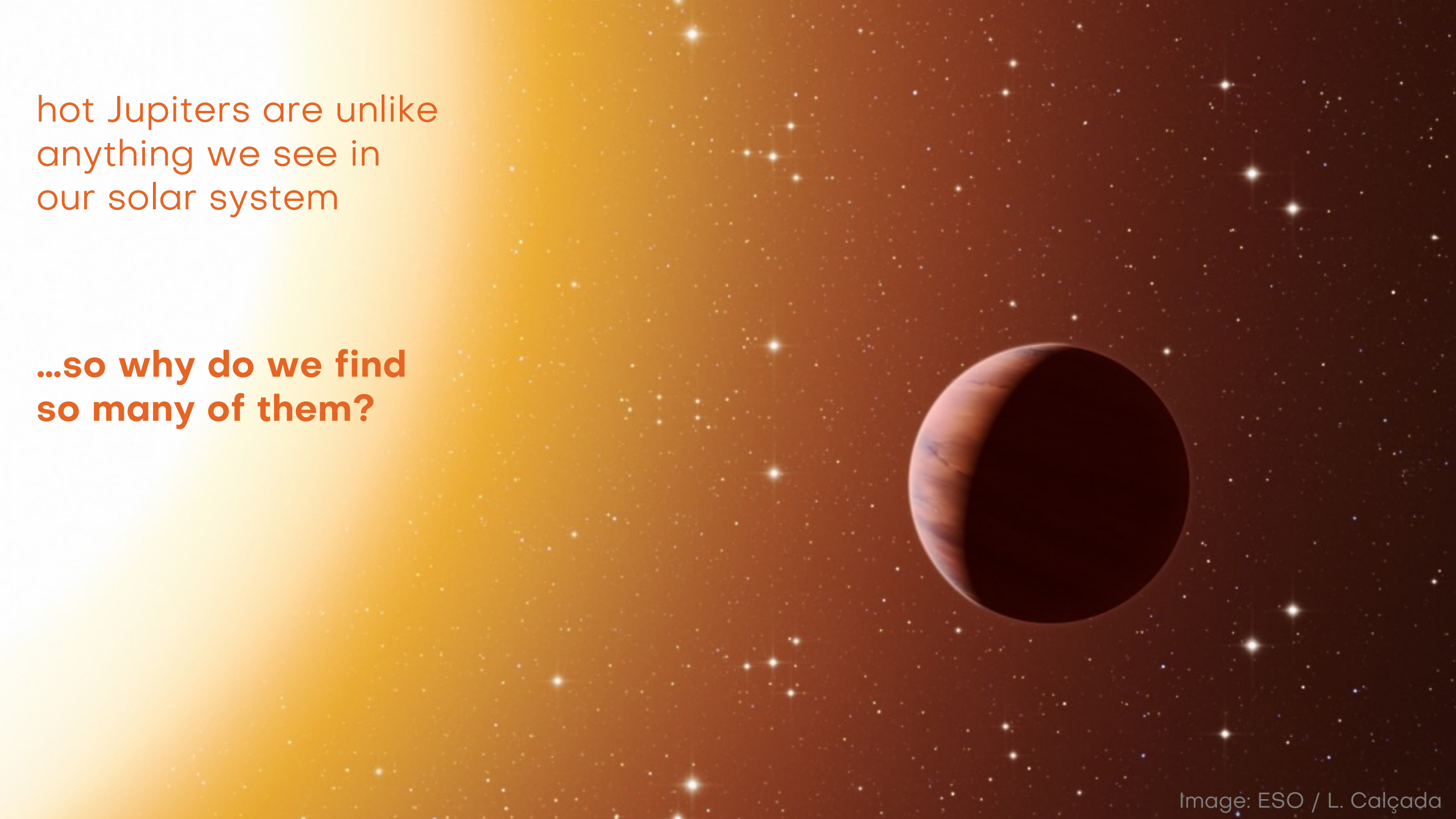


Image: ESO / L. Calçada

Are hot Jupiters common?

there are many in our galaxy, but they're not as common as they appear on this plot

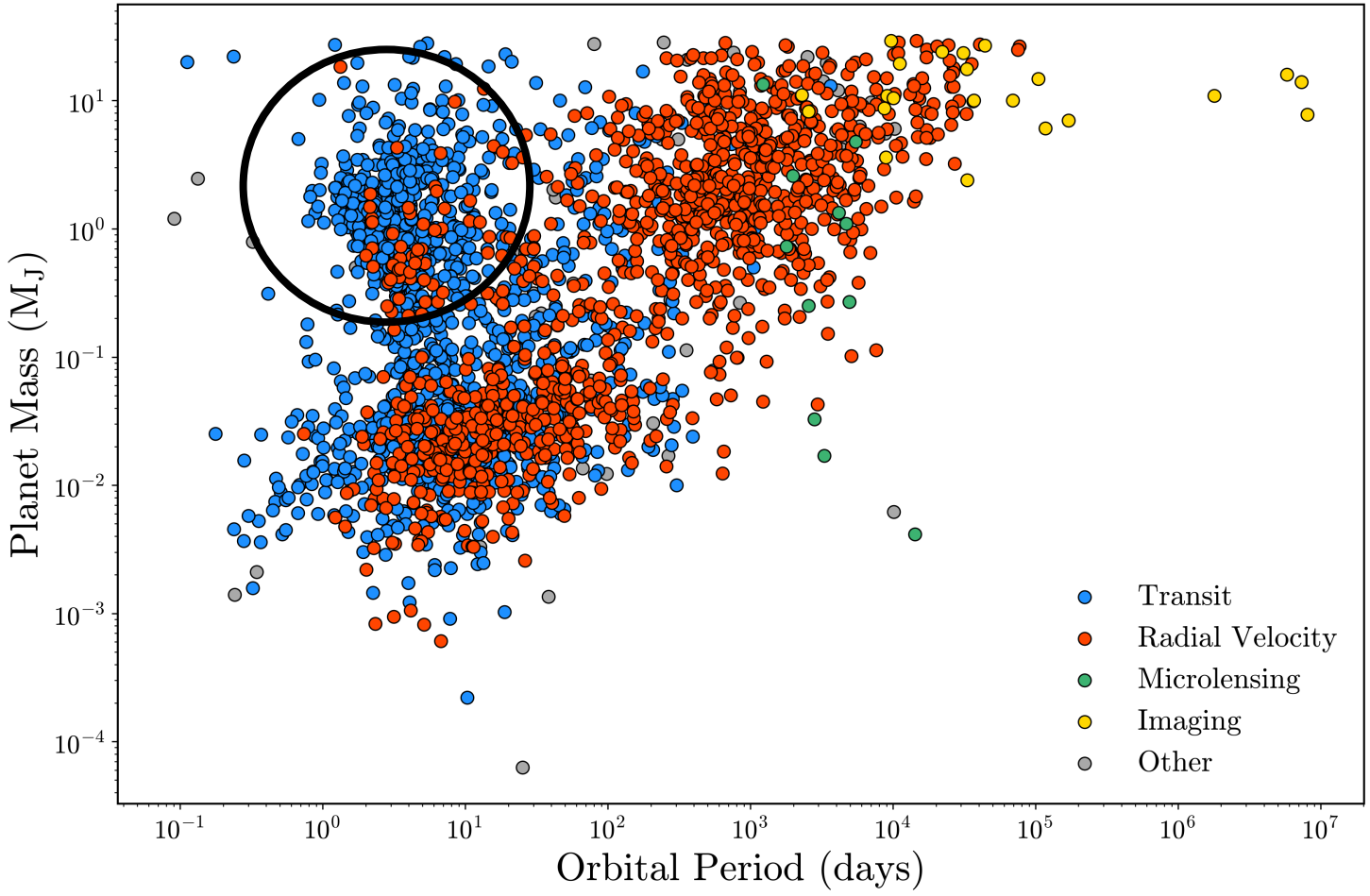




Image: P. Schmidt (Terranaut)

Tutorial: Recovering a transiting planet in TESS data

Step 0:
⟶ Install lightkurve & plot a sample light curve

Step 1:
⟶ Explore the NASA Exoplanet Archive, then make cuts to choose your planet
⟶ Download & plot its TESS light curve

Step 2:
⟶ Find the orbital period using a Box Least Squares periodogram
⟶ Fold the light curve to this period & plot

Step 3:
⟶ Tune the transit depth parameter to estimate the planet radius*
**Requires the star's radius! Look it up on the archive*

Step 4:
⟶ Compare your period & radius to the Exoplanet Archive values
⟶ Plot your planet in context, then submit your results to the class plot

Link: ***nksaunders.space/tutorials***
Make sure to click File → Save a copy in Drive